



ECON 100B: Macroeconomics

Class 18: Monetary Policy & the Phillips Curve [1/2] **(Jones Chapter 12)**

Ryan D. Edwards
April 2, 2026

Agenda

- Finish the IS curve and the multiplier, from Chapter 11
- Chapter 12: Monetary policy and the Phillips curve

Announcements

- **Problem Set 3** is available, due on **Friday, April 17**
- **Problem Set 4** will be due on **Friday, May 1**
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in Pimentel 1
 - We will run 90 minutes from about 3:00-4:30PM
 - Like at midterm, DSP students please either make a proctoring appointment, or for 150% time only, we will get another classroom

April is Arab American Heritage Month

- [UC Berkeley page](#)
- Notable: [2025 Nobel Prize in Chemistry](#) awarded in part to another Berkeley chemist:

James and Neeltje Tretter Prof. [Omar Yaghi](#)

- Born in 1965 in Amman, Jordan
 - In 9th grade around 1980, his father instructed him to get a student visa to study in the U.S., and to take the entire family savings (\$9K) with him
 - Nobel-winning research on “metal-organic framework” capable of holding many more molecules inside them
- (Recall the [2020 Nobel Prize in Chemistry](#) went in part to Berkeley Prof. Jennifer Doudna for gene therapy CRISPR)

Learning objectives for Chapter 12

- The full short-run model consists of
 - A downward-sloping **IS curve**
 - A horizontal **MP curve** at the interest rate R chosen by the central bank
 - The **Phillips curve**, an upward-sloping relationship between $\Delta\pi$ and $\tilde{Y}$
- External price shocks can raise or lower the Phillips curve
- To reduce inflation through the Phillips curve, the central bank must cause a (short?) recession with $\tilde{Y} < 0$
- The Great Inflation disaster of the 1970s in the U.S. was partly due to monetary policy that was too “loose,” meaning R was too low

These are **NOT** learning objectives for ECON 100B

- How monetary policy is actually conducted by the Fed and other central banks
 - The “nuts and bolts” have been in flux over the decades
 - But the bottom line is the same:

The Fed and other central banks **set the short-run real interest rate**
(with some constraints)

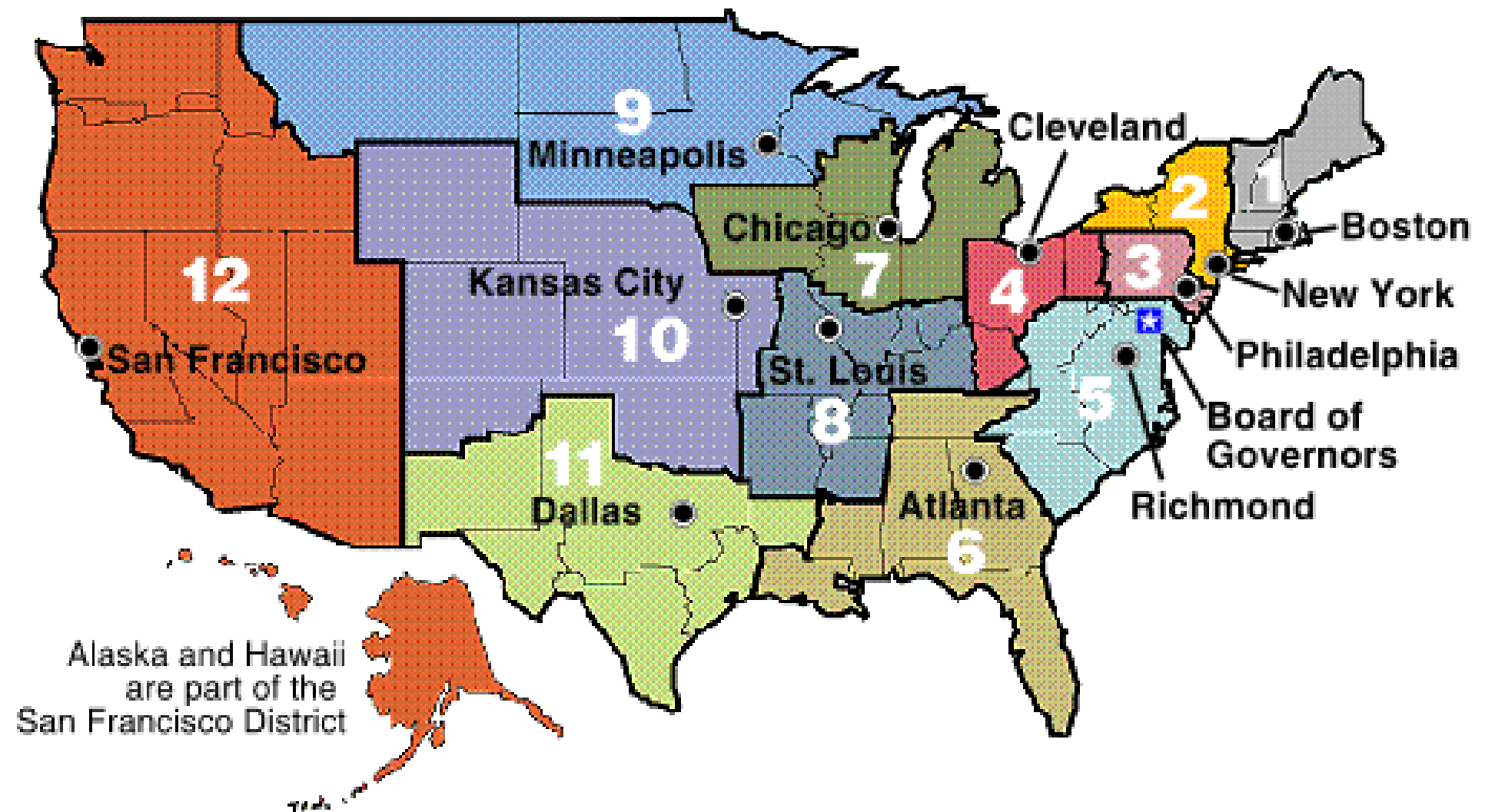
- The Fed’s balance sheet, quantitative easing

The multiplier

Slides 35-53 of Class 17

What is the Federal Reserve?

- It is a system of **12 regional banks**
- The NY Fed is the largest in terms of assets
- Plus a **Board of Governors** in Washington, DC
- U.S. banking has a long, odd history dating back to Alexander Hamilton



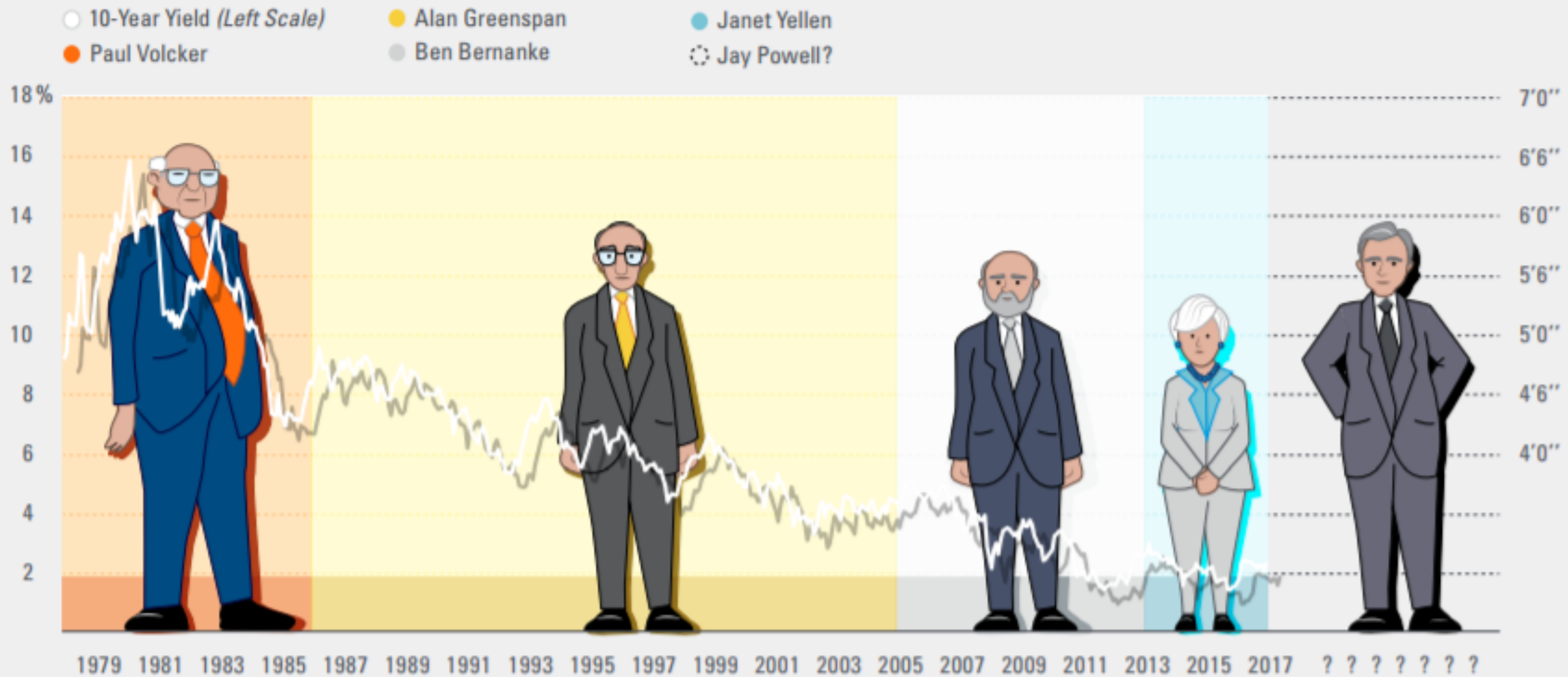
- In 1913, Woodrow Wilson signed the Federal Reserve Act into law, which “provided for the establishment of Federal Reserve Banks, to furnish an elastic currency, to afford means of rediscounting commercial paper, *to establish a more effective supervision of banking* in the United States, and for other purposes.”

A History of the Fed Funds Rate Since 1979



Source: Bob Rich from Hedgeye.

BOTH THE HEIGHT OF THE FED CHAIR AND RATES HAVE FALLEN OVER TIME, COULD A TALLER FED CHAIR MEAN RATES RISE?



Source: LPL Research, Bloomberg 10/22/17

We don't actually believe that interest rates are determined by the height of the Fed chair, but it has been an interesting coincidence.

What's up with the [Police Lineup?](#)

Source: [CNBC citing LPL Financial](#)

Quoted at the beginning of Jones Chapter 12

Remarks at Chair Janet L. Yellen's Ceremonial Swearing-in, March 5, 2014

Two statutory objectives of the Federal Reserve System

- In 1978, Jimmy Carter signed the Humphrey-Hawkins Full Employment and Balanced Growth Act
- Humphrey-Hawkins directs the Federal Reserve to control inflation and promote full employment
- In her remarks, Janet Yellen described this as:

“The goals set by Congress for the Federal Reserve are clear: **maximum employment** and **stable prices**.”

- Formally, we might say:
 - “Stable prices” or “price stability” usually means “low and stable inflation”
 - “Maximum employment” usually means “a low and stable unemployment rate,” or hiccups in $\tilde{Y}$ that are temporary

Get ready for a mind-bender

- The Fed is bound to the twin mandate by statute ...
- But in ECON 100B, we will start off by assuming that the Fed *only has an inflation target $\bar{\pi} = 0.02$*
- The European Central Bank, for example, only has the formal objective of “price stability,” as do many other central banks
- But it turns out that the nature of the **Phillips curve** means that maintaining $\bar{\pi} = 0.02$ *also means maintaining $\tilde{Y} = 0$*
- The kicker is that sometimes, inflation in globally traded goods can spread from world markets and raise π while lowering $\tilde{Y}$

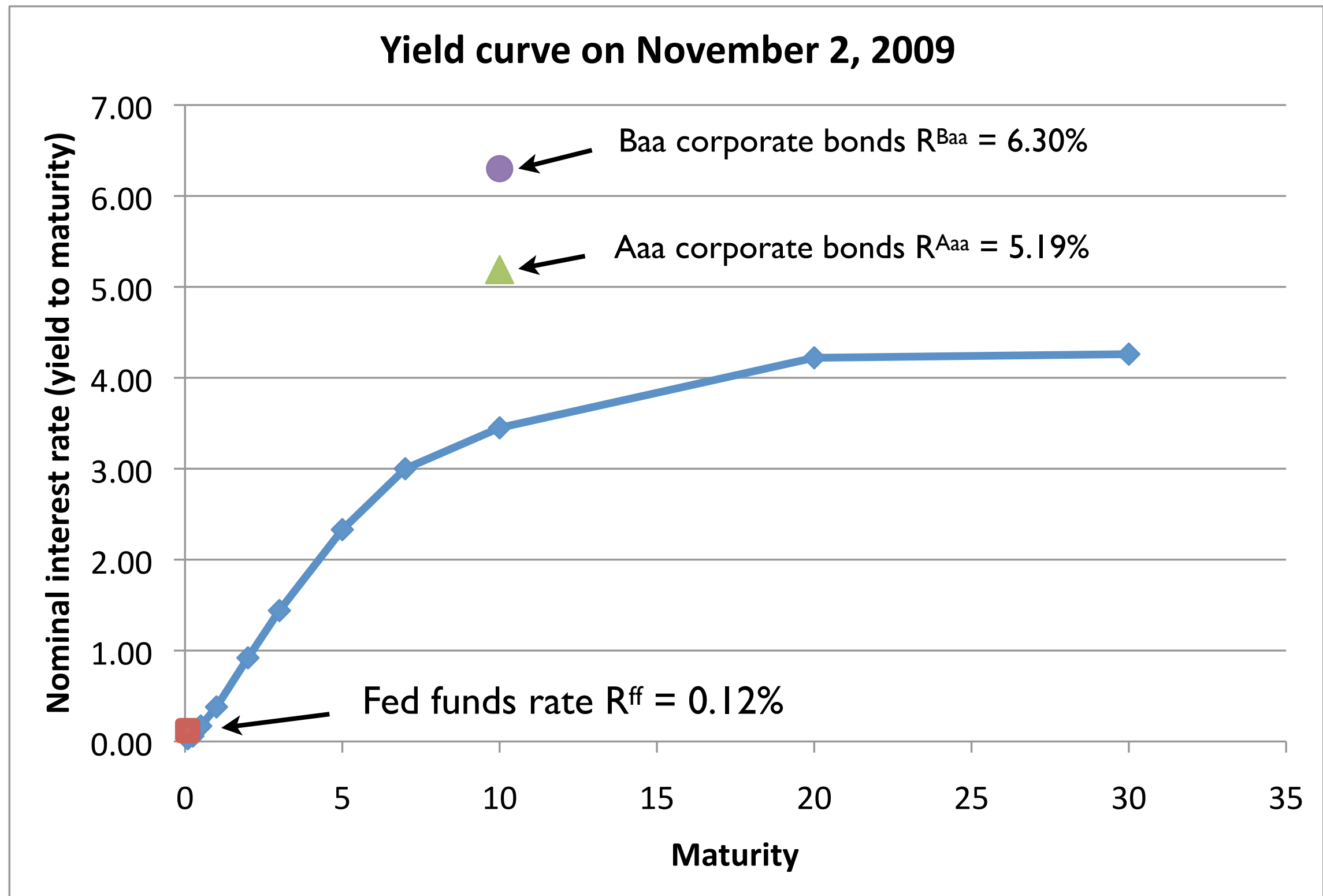
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The federal funds rate R^{ff} and other interest rates

- The federal funds rate R^{ff} is the rate that banks charge each other on overnight loans
- It is a **short-term** interest rate
- The Fed can effectively set the value of R^{ff} because banks could also borrow from the Fed through the discount window
- Businesses that invest in capital typically require loans with much longer terms or “maturities”
- Banks typically charge **term premiums** on longer loans. They also charge **risk premiums** when the borrower is riskier

The Treasury yield curve shows term premiums.
We can also see risk premiums in 10-year corporate bonds



Technically the Fed affects *nominal interest rates*. What about the real interest rate?

- We know that the nominal interest rate, i_t , is related to the real interest rate, R_t , through the Fisher equation:
- Does inflation change? Sure, but *not in the short run*
- This will be a key assumption of ours, and it's basically right: inflation will not change a lot over short periods of time
- So changes in the nominal federal funds rate will typically produce changes in the real federal funds rate, R^{ff}

Technically the Fed sets *nominal interest rates*. What about the real interest rate?

- The nominal interest rate i is related to the real interest rate R and inflation through the Fisher equation:

$$R = i - \pi$$

- Does inflation change? Sure, but if inflation is under control, it doesn't change much in the short run
- Changes in the nominal federal funds rate will typically produce changes in the real federal funds rate, R^{ff}

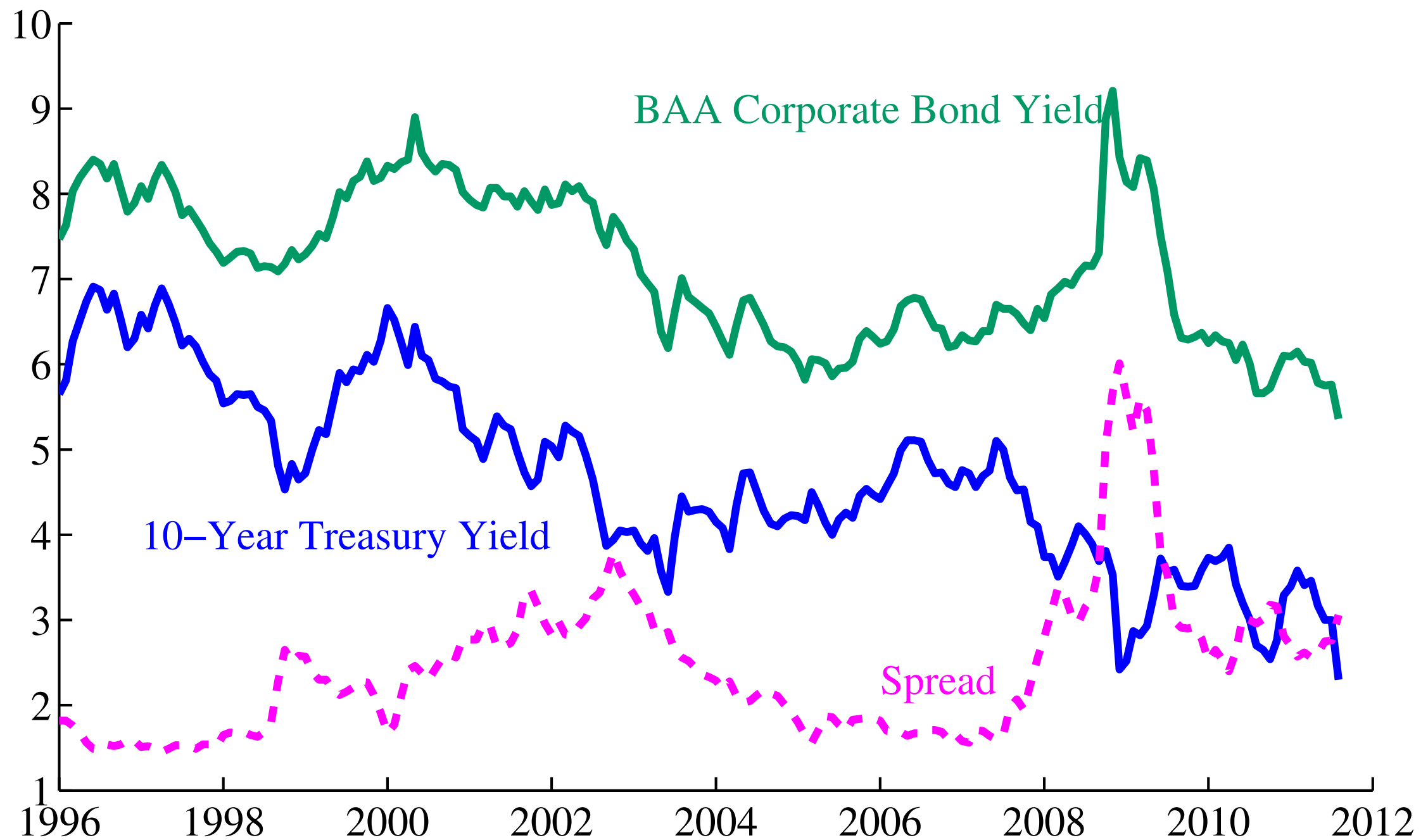
The Fed sets the federal funds rate, R^{ff} , but markets set the risk premium

- The Fed can choose R^{ff}
- But then financial markets decide what risk premium $\bar{p}$ they need to charge on business loans
- (Term premiums matter, but they are more predictable than risk premiums, which can really wreak havoc)
- In our model, we'll assume there is one “average” interest rate R that is important for businesses in the IS curve

$$R = R^{ff} + \bar{p}$$

- During normal times, we'll assume that $\bar{p} = 0$

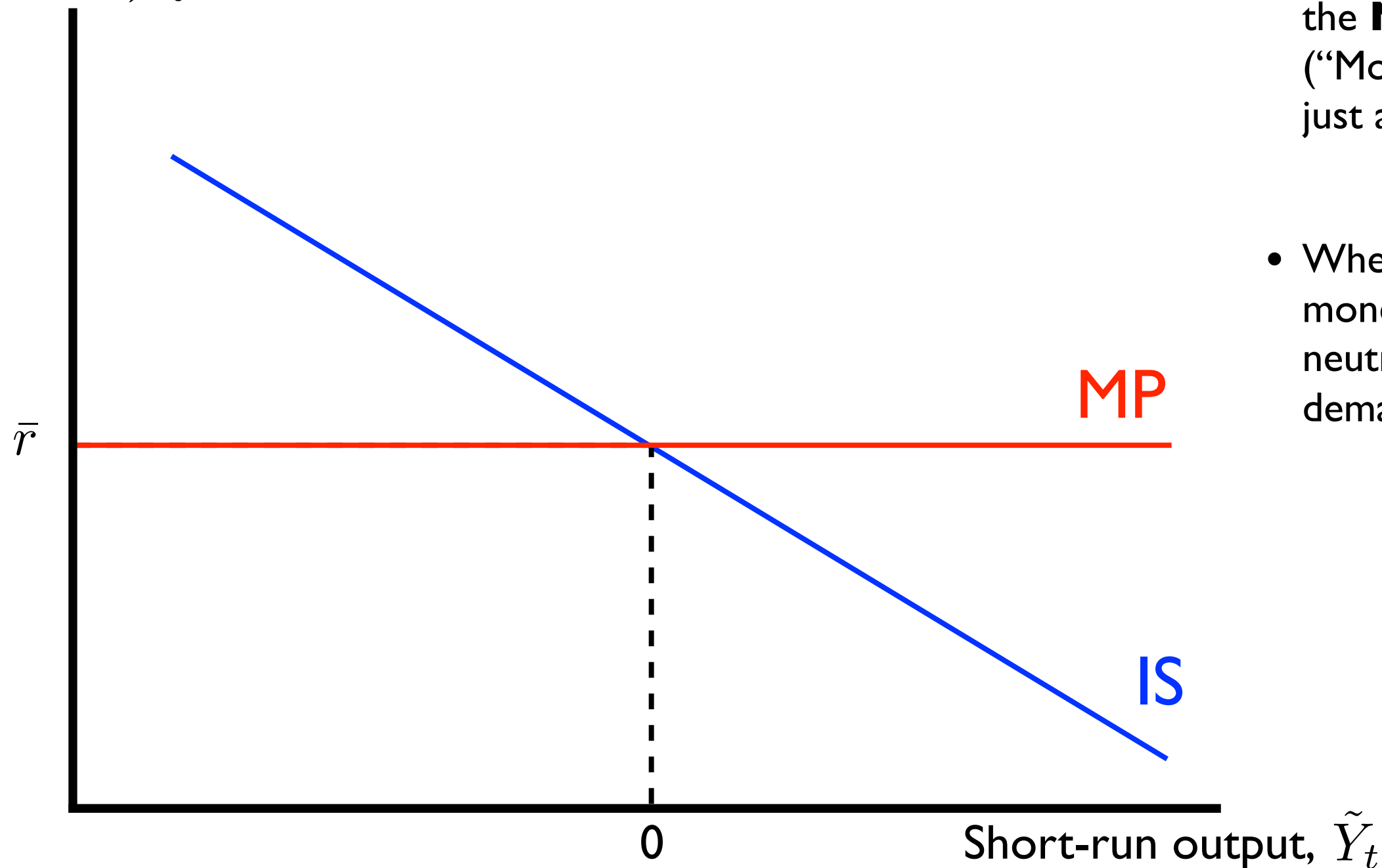
Times have notably *not* been normal



The “spread” between Baa corporate bonds and the 10-year Treasury, which is the risk premium $\bar{p}$, increased dramatically in 2008

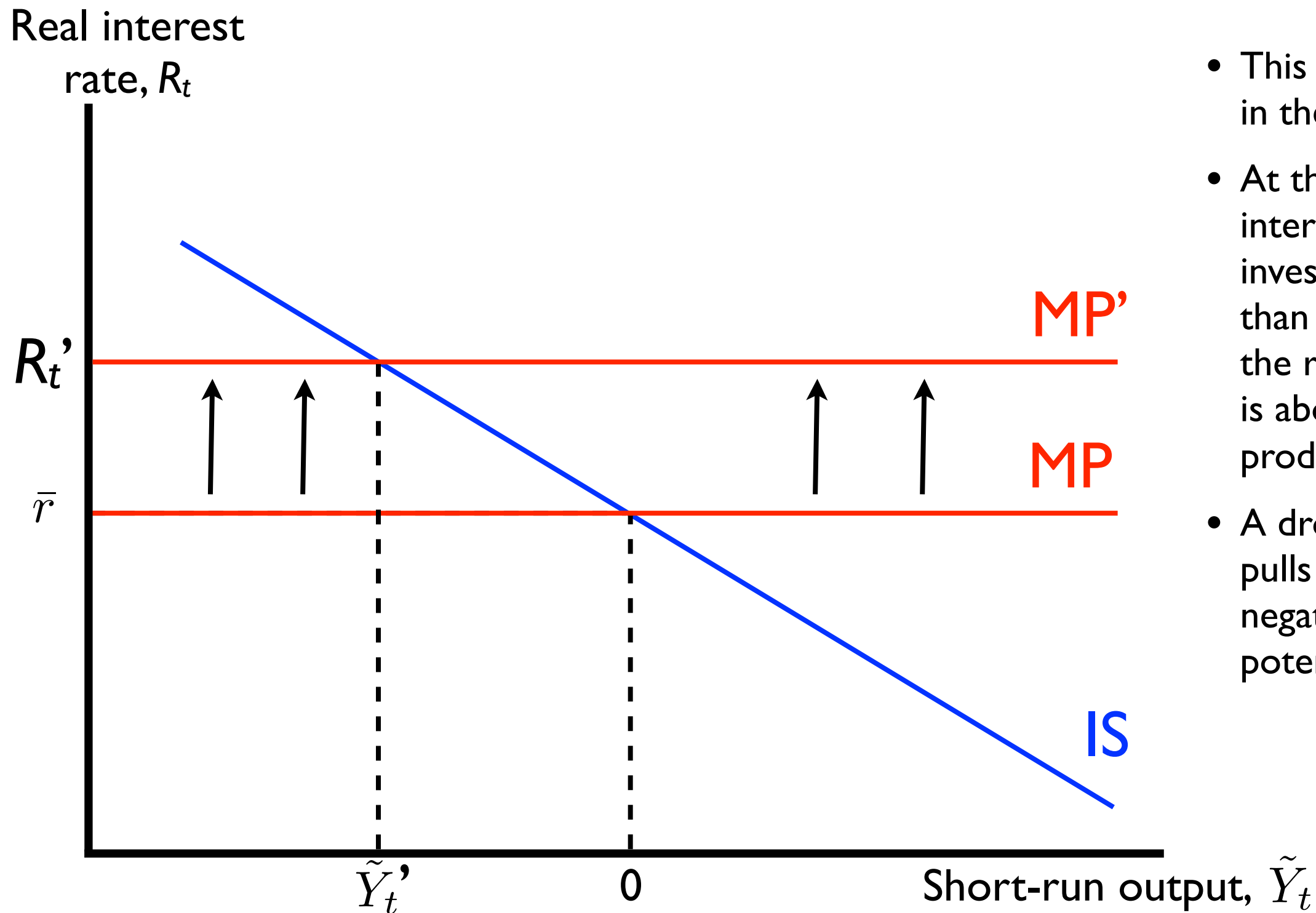
If the Fed can determine the real interest rate in the short run, then here's our new short-run model diagram:

Real interest
rate, R_t



- The Fed can set the real interest rate, so the **MP Curve** (“Monetary Policy”) is just a horizontal line
- When $R = \bar{r}$, monetary policy is neutral. Without any demand shocks, $\tilde{Y} = 0$

Suppose the Fed raises the real interest rate

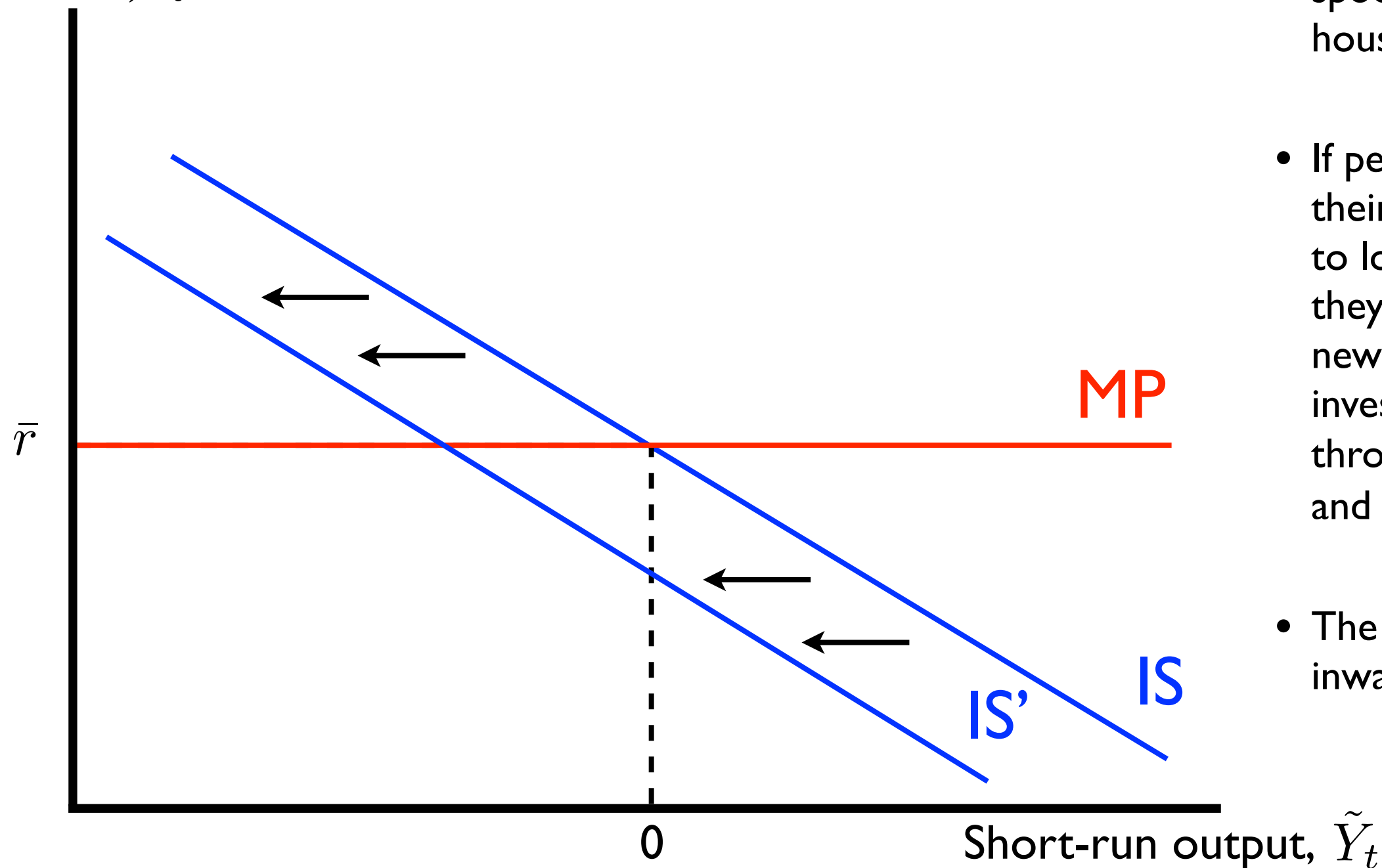


- This is an upward shift in the MP curve
- At the higher real interest rate R_t' , investment is lower than before, because the real interest rate is above the marginal product of capital, $\bar{r}$
- A drop in investment pulls short-run output negative, below potential output, to

$$\tilde{Y}_t' < 0$$

Why might the Fed need to act? Consider a bursting housing bubble and aggregate demand

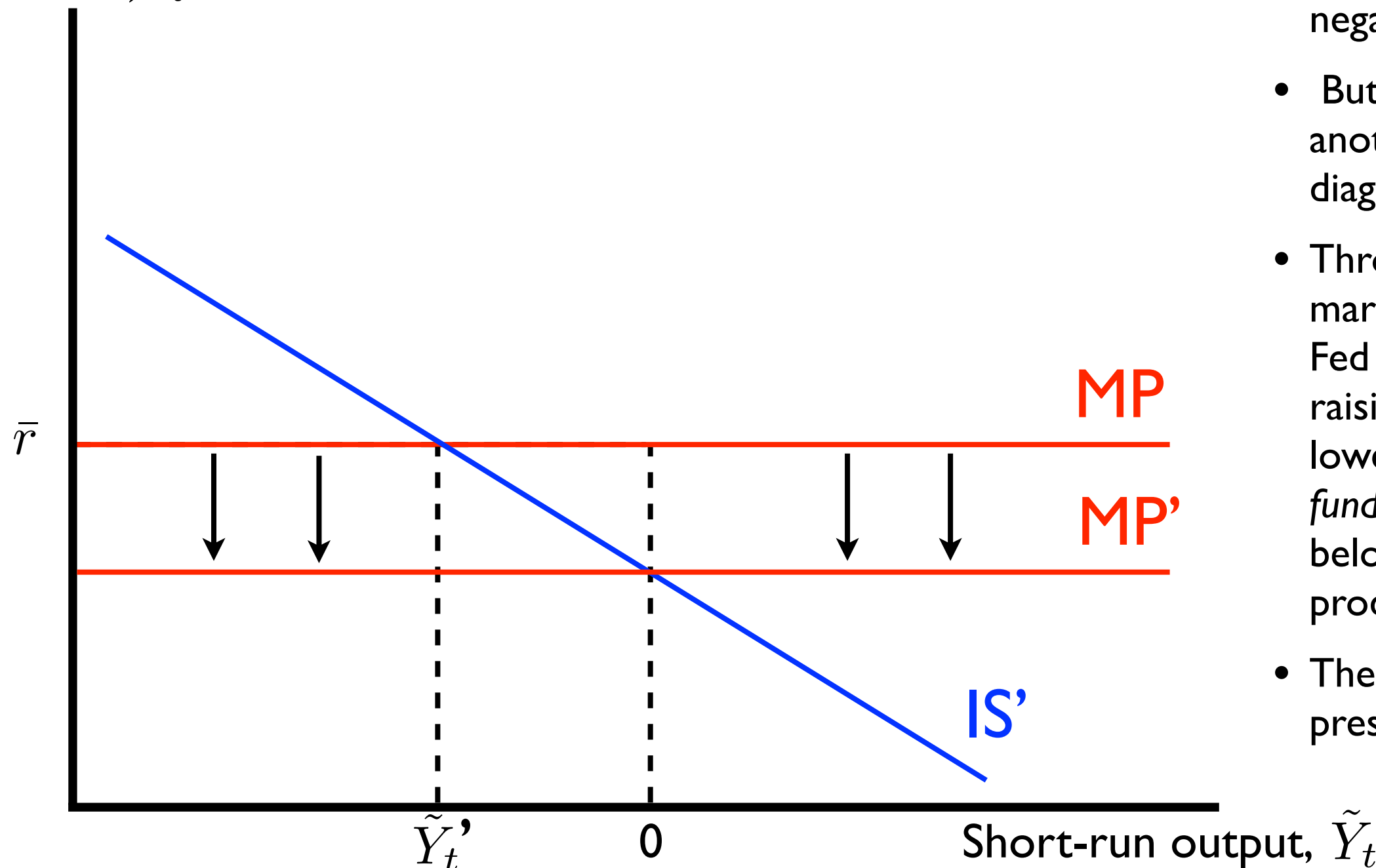
Real interest rate, R_t



- Suppose people get spooked by the housing market
- If people fear that their housing is about to lose value quickly, they might stop buying new housing, shifting investment down through a decline in $\bar{a}_i$ and in $\bar{a}$
- The IS Curve shifts inward to the left

Without action, short-run output will decline

Real interest
rate, R_t



- The IS curve has fallen, to IS' , which will pull short-run output negative, to $\tilde{Y}_t' < 0$
- But now, there's another curve in this diagram!
- Through its open market window, the Fed buys bonds, raising their prices and lowering the *federal funds rate*, pushing R_t below the marginal product of capital,
- The MP curve falls and preserves

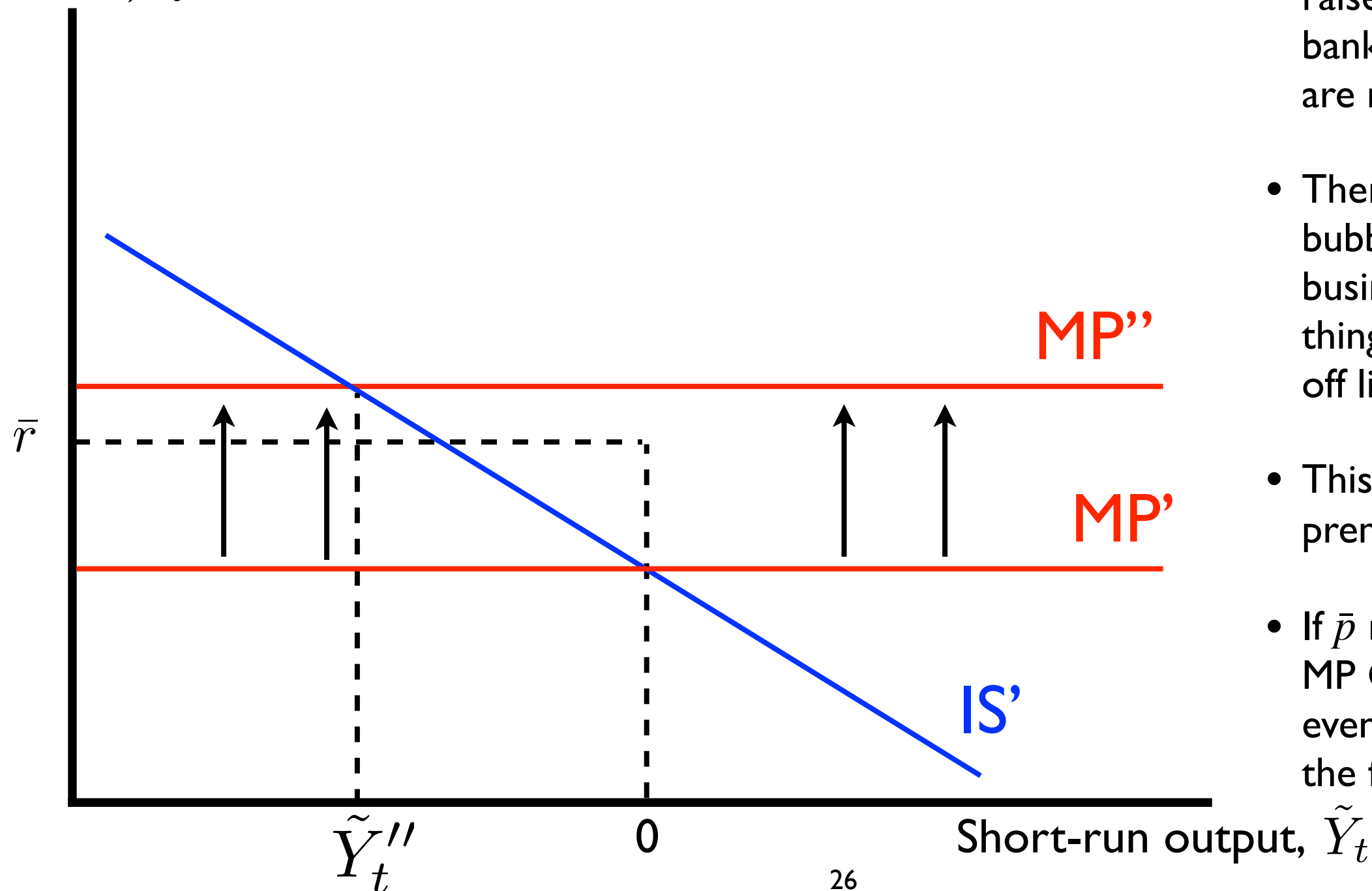
$$\tilde{Y}_t = 0$$

A problem arises when perceptions of risk rise!

Real interest
rate, R_t

$$R_t = R^{ff} + \bar{p}$$

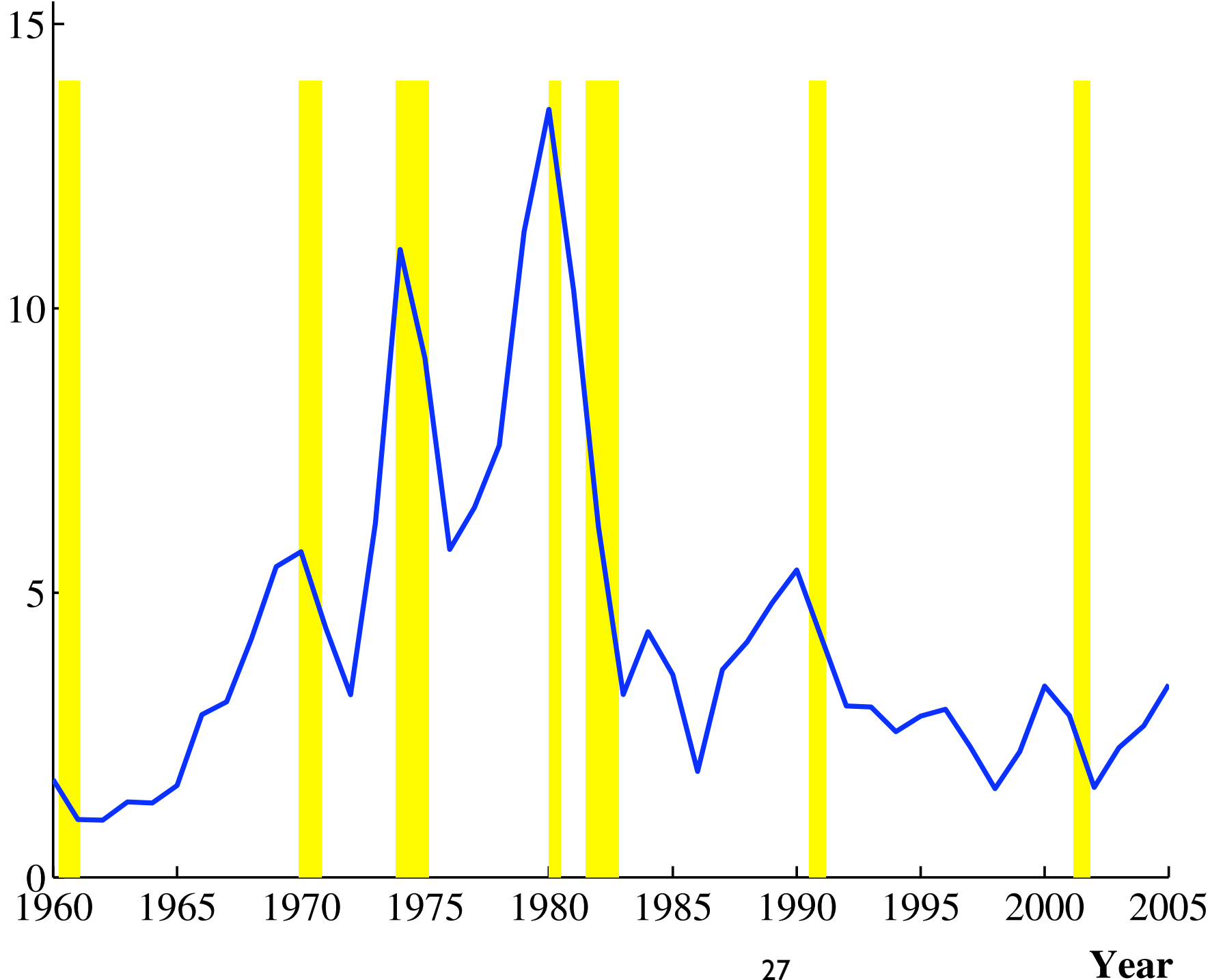
The MP Curve



- Suppose the bursting of the housing bubble raises fears among banks that their loans are now riskier. Why?
- There was an asset bubble, which means businesses did dumb things that didn't pay off like they thought
- This raises the risk premium $\bar{p}$
- If $\bar{p}$ rises enough, the MP Curve *might* rise even after the Fed cut the fed funds rate, R^{ff}

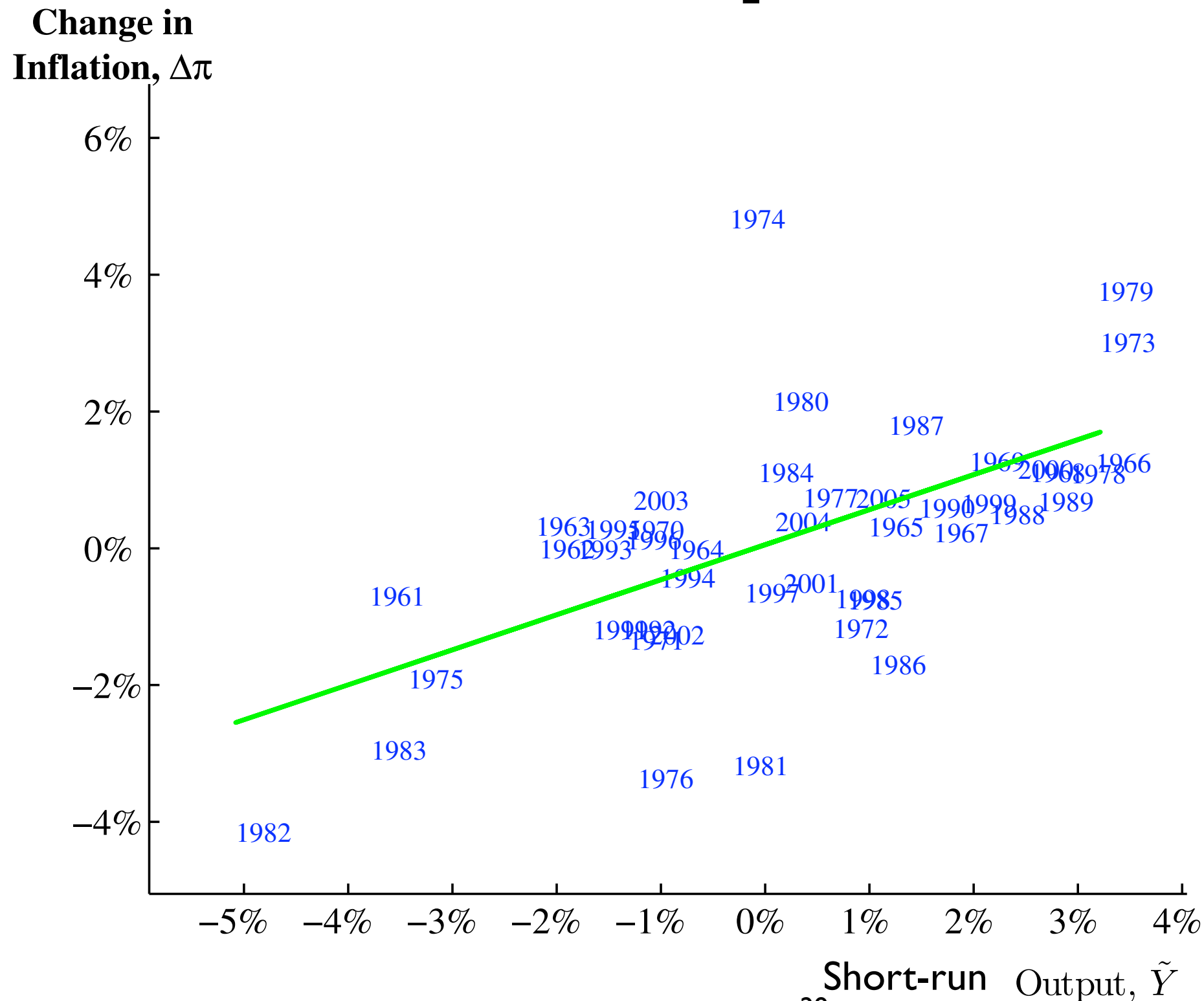
Next: We have seen how **inflation** frequently rises during expansions and almost always falls during recessions

Inflation
(Percent)



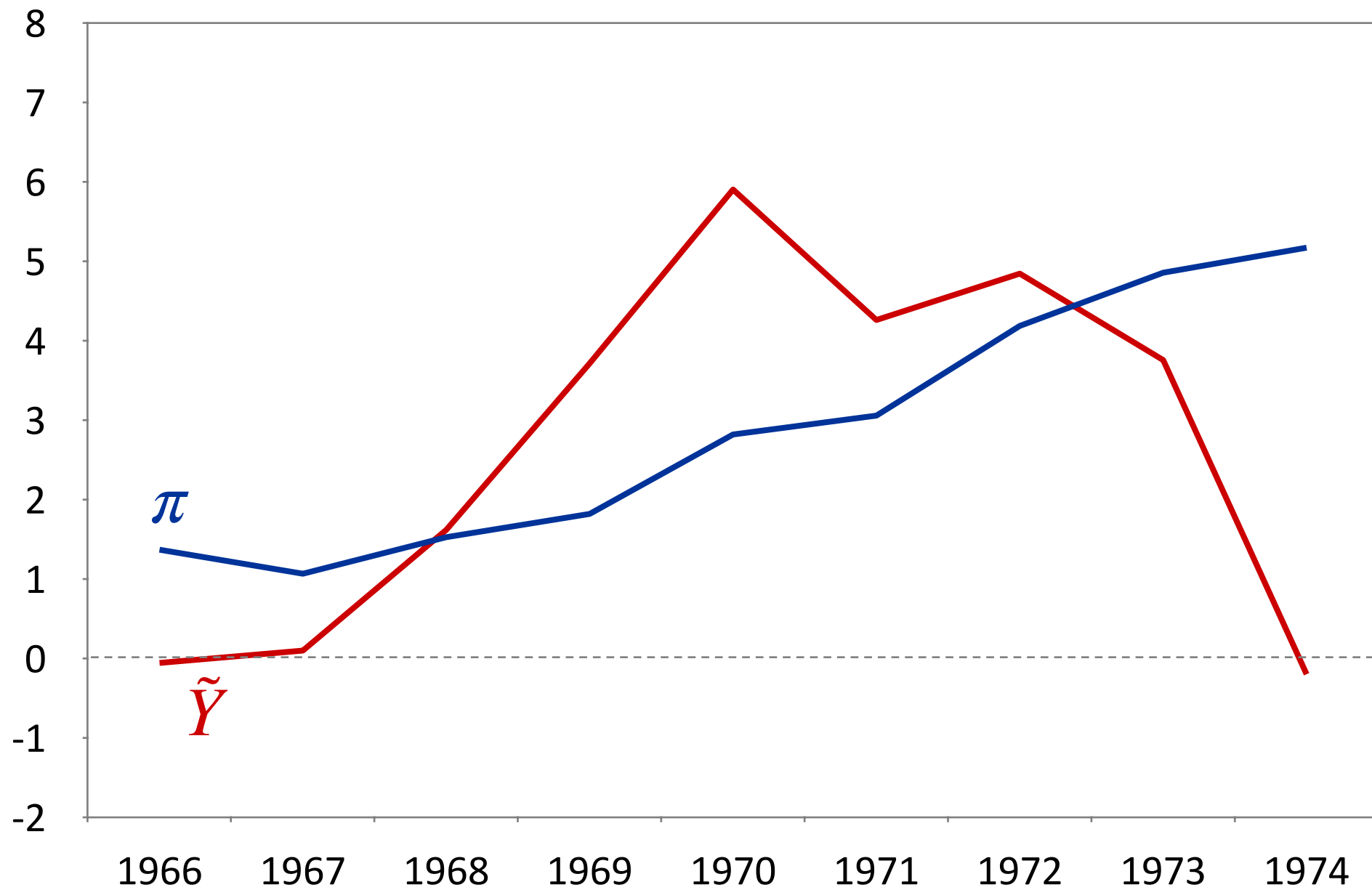
- We know that in the **long run**, inflation is tied to the growth rate of the **money supply** through the quantity theory of money: $MV = PY$
- But inflation in the **short run** is associated with booms and busts

Earlier, we learned about the **Phillips curve**



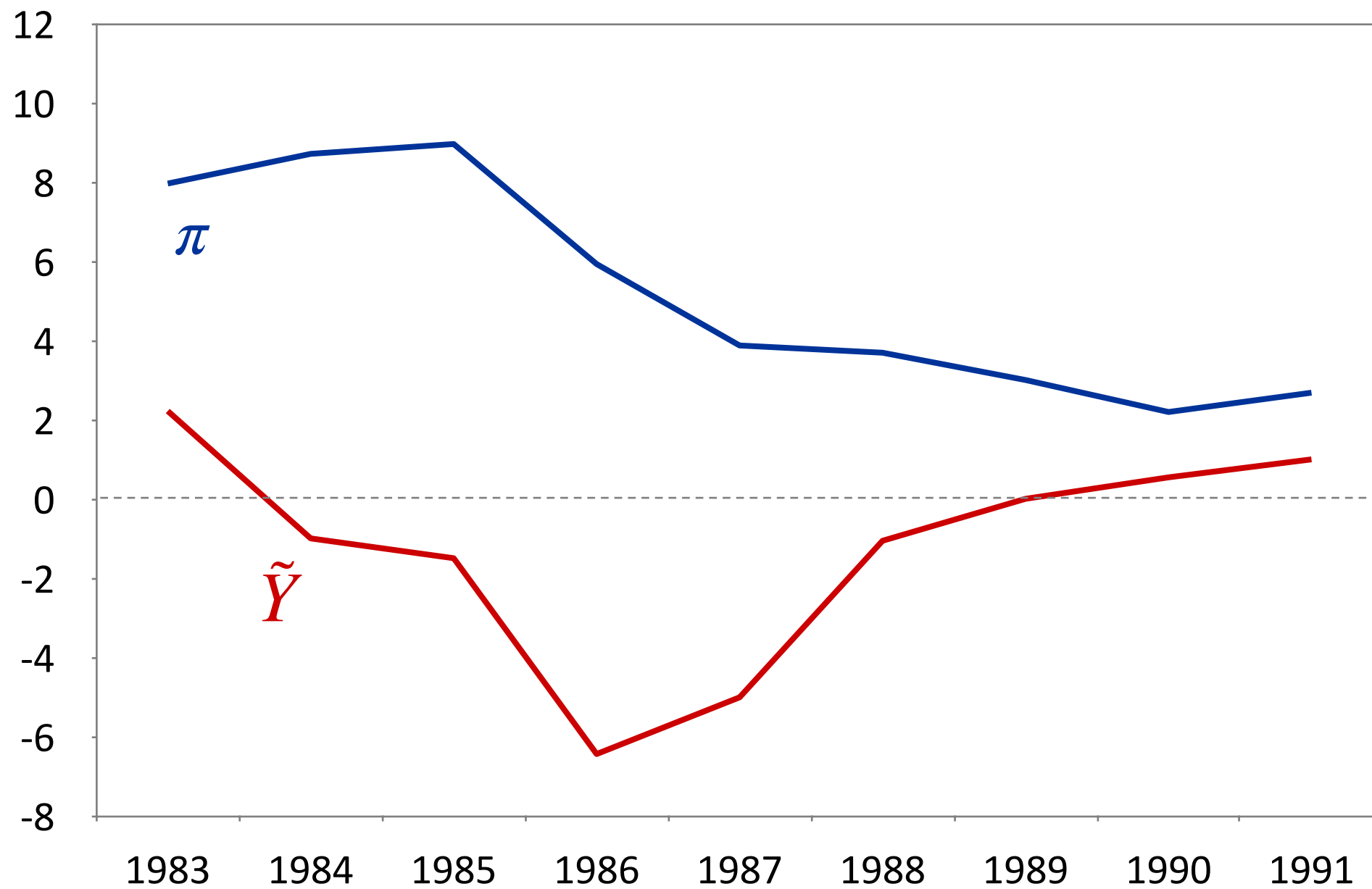
- If you plot short-run output on the horizontal axis versus the change in inflation on the vertical axis, you see an upward-sloping relationship, *but with noise*
- This summarizes a key point: when output is *temporarily* higher, there are inflationary pressures: Furby prices rise when demand is high
- (But more inflation cannot produce higher output in the long run)

Inflation and Output, 1962–1970



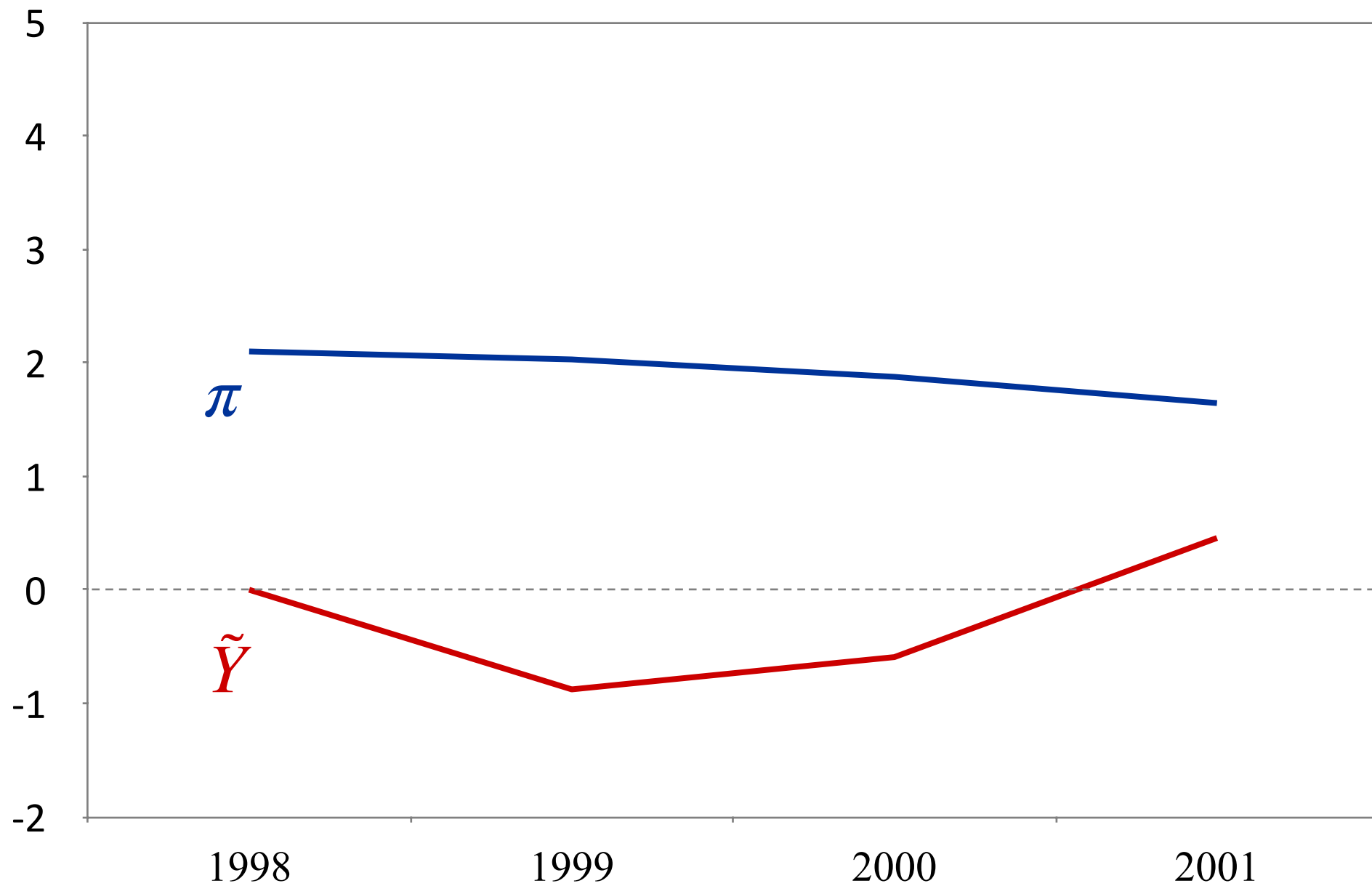
Source: Bureau of Economic Analysis.

Inflation and Output, 1979–1987



Source: Bureau of Economic Analysis.

Inflation and Output, 1994–1997



Source: Bureau of Economic Analysis.

How does inflation happen?

Firms set their prices

- The money supply *must* determine the price level in the long run, but in the short run, the action is with firms and prices
- Imagine a store that produces and sells ice cream
- For the last several years, it has noticed that all prices are rising at 3% each year, so $\pi = 0.03$ usually
- But suppose the owner of the ice cream store reads in the paper that many local workers just lost their jobs — and the local workers are the folks who usually eat ice cream
- The owner might lay off some workers and supply less ice cream, but he/she will probably raise prices less than 3%

What does this price-setting behavior mean for our model?

- Actual economy-wide inflation will be equal to what the firm owners *think* economy-wide inflation will be ...
 - Minus something when owners expect a slump, or
 - Plus something when owners expect a boom

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}_t$$

Actual inflation

Expected inflation

an increasing
function of
short-run output

Suppose $\bar{v} = 0.5$ and $\pi_t^e = 0.03$. Then when output is 1% above potential, $\tilde{Y}_t = 0.01$ and inflation will be higher too:

$$\pi_t = 0.03 + 0.5(0.01) = 0.035$$

But what are expectations about inflation — π_t^e ?

- Sometimes they can really get out of control, which they did during the 1970s
- If firms *expect* high inflation, then it's basically a foregone conclusion! High inflation will simply happen because:

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}_t$$

- But suppose inflation is relatively subdued. Then firms will expect inflation to be what it has been in the recent past:

$$\pi_t^e = \pi_{t-1}$$

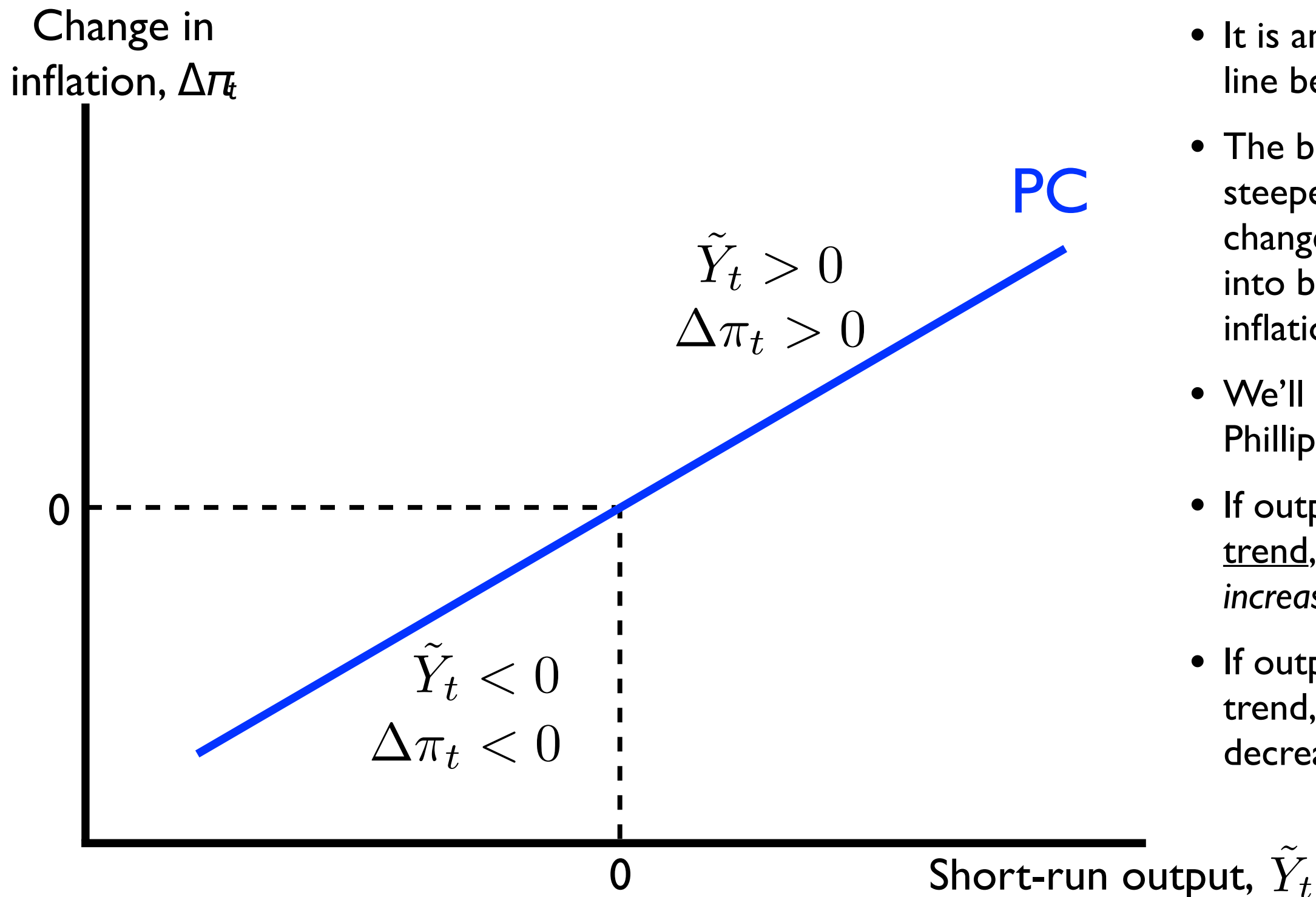
- Under these “adaptive expectations,” we find that

$$\pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t \quad \text{so}$$

$$\Delta\pi_t = \bar{v}\tilde{Y}_t$$

This equation is the Phillips Curve

$$\Delta\pi_t = \bar{v}\tilde{Y}_t$$

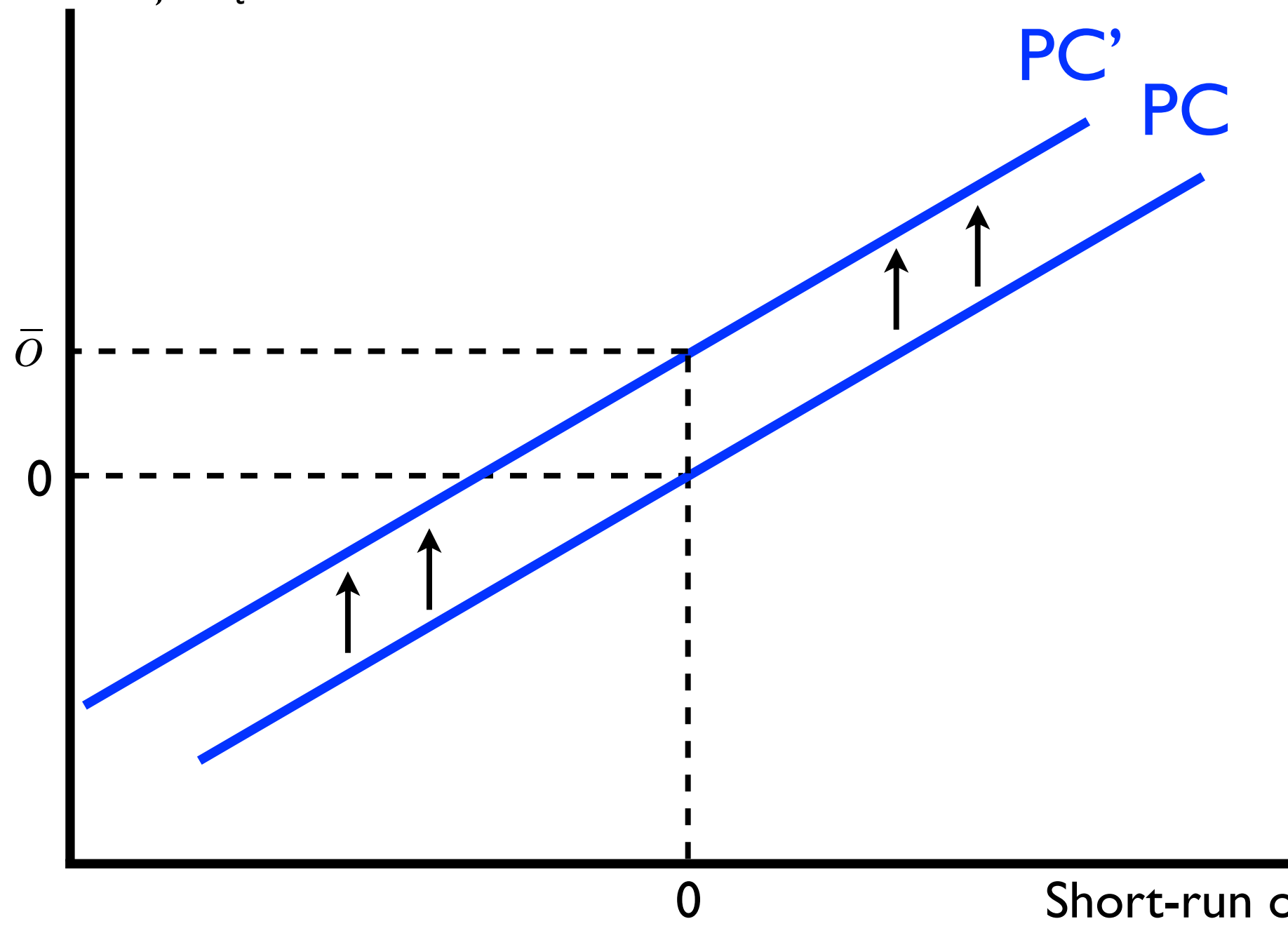


- It is an upward sloping line because the slope, $\bar{v} > 0$
- The bigger $\bar{v}$ is, the steeper the curve, and changes in output feed into bigger changes in inflation
- We'll call this "PC" for Phillips Curve
- If output is above trend, then inflation *increases*
- If output is below trend, then inflation *decreases*

Oil shocks and other events can raise inflation

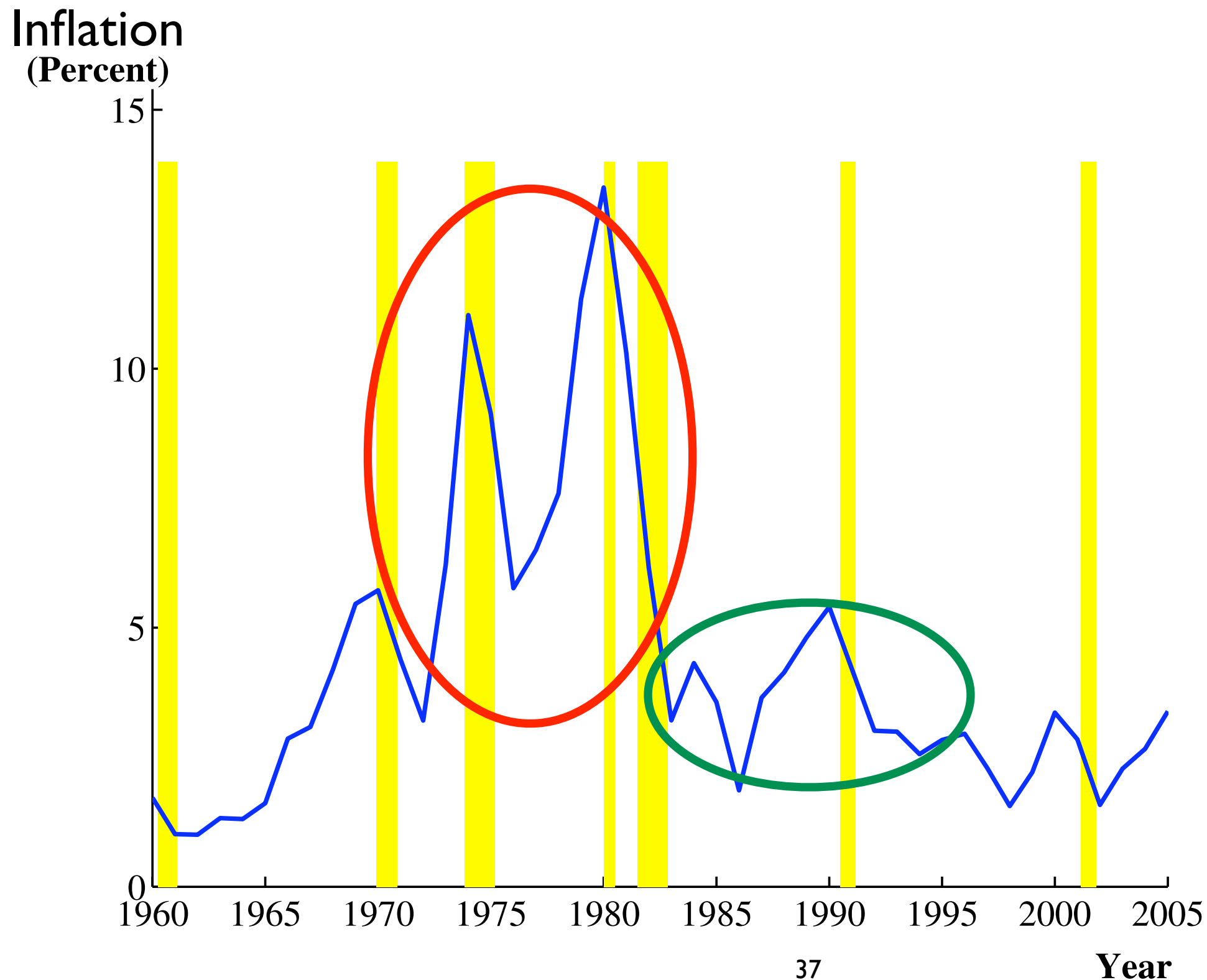
$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$

Change in
inflation, $\Delta\pi_t$



- Suppose the supply of oil is restricted by the OPEC production cartel, and the price of oil rises
- Oil is an input in the production of many goods and services, so many producers have to raise prices!
- We will model the oil shock as an additive shift to $\Delta\pi_t$, raising inflation at every level of output
- This shifts the PC curve upward
- At potential output, inflation increases!

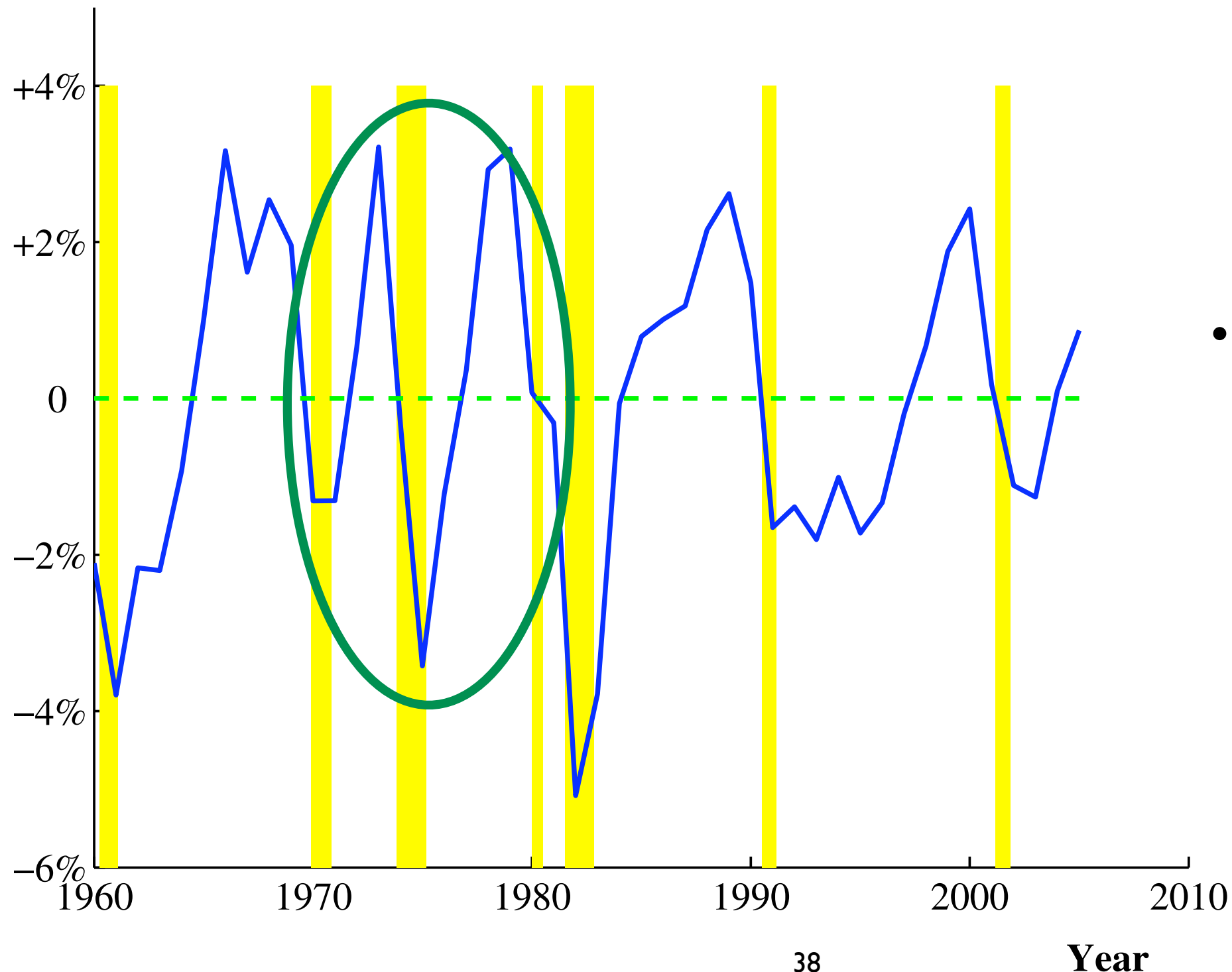
We'll use the full short-run model to explain macroeconomic history in the 1970s and 1980s



- **First**, why did inflation get to be such a problem in the 1970s, especially by 1980?
- **Second**, how did policymakers regain control over inflation and bring it back down to levels not seen since the 1960s?
- (The textbook covers these in reverse, but I'll cover them chronologically)

Why did inflation increase in the 1970s?

Short-Run Output, $\tilde{Y}$



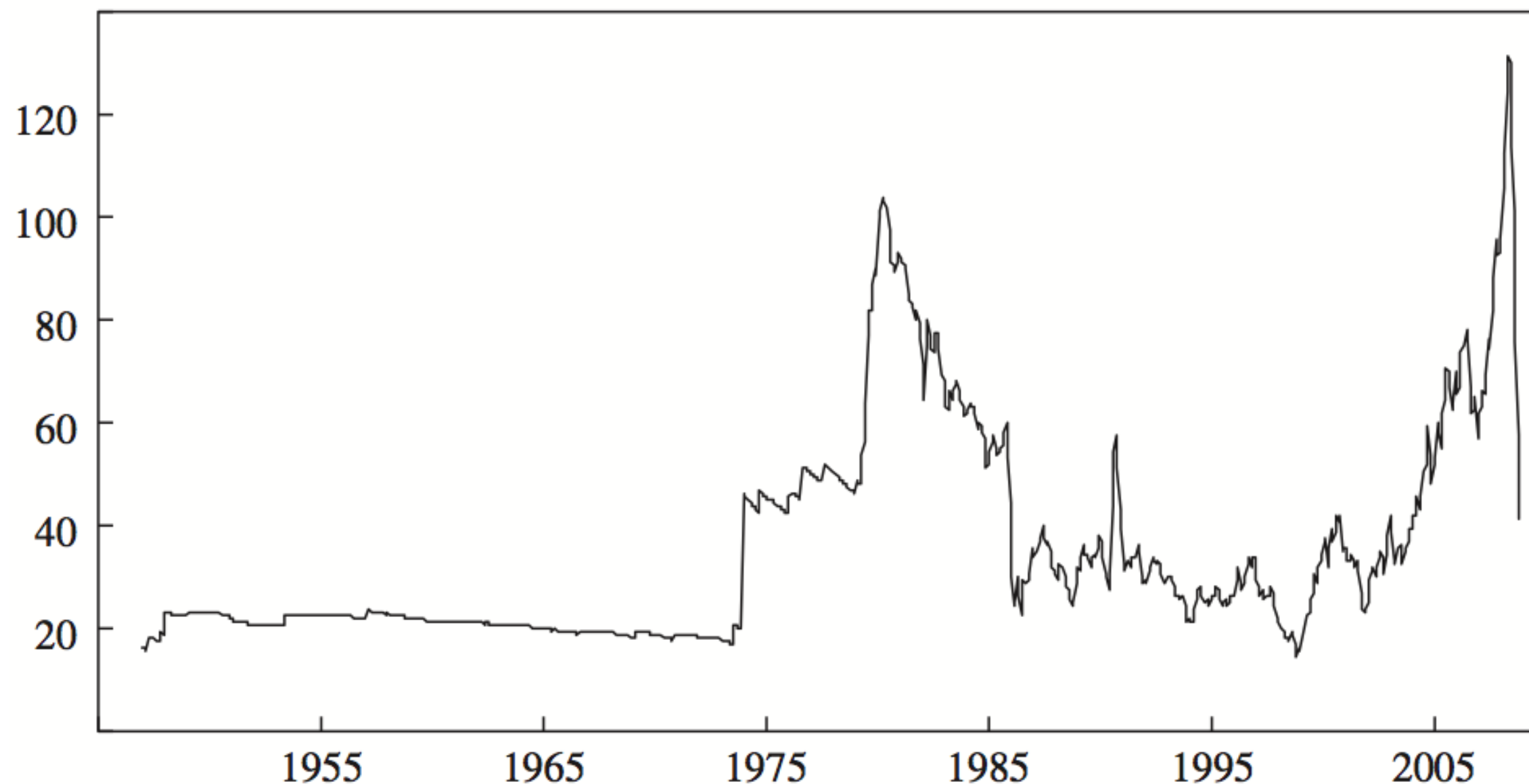
- Short-run output was swinging back and forth in a pretty normal fashion — there really wasn't a massive inflationary expansion
- So the high inflation was **not** the result of any movement along the Phillips Curve, because there really wasn't any permanent change in short-run output

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$

But shocks definitely raised oil prices in the 1970s

Figure 1. Real Crude Oil Prices, January 1947 to December 2008^a

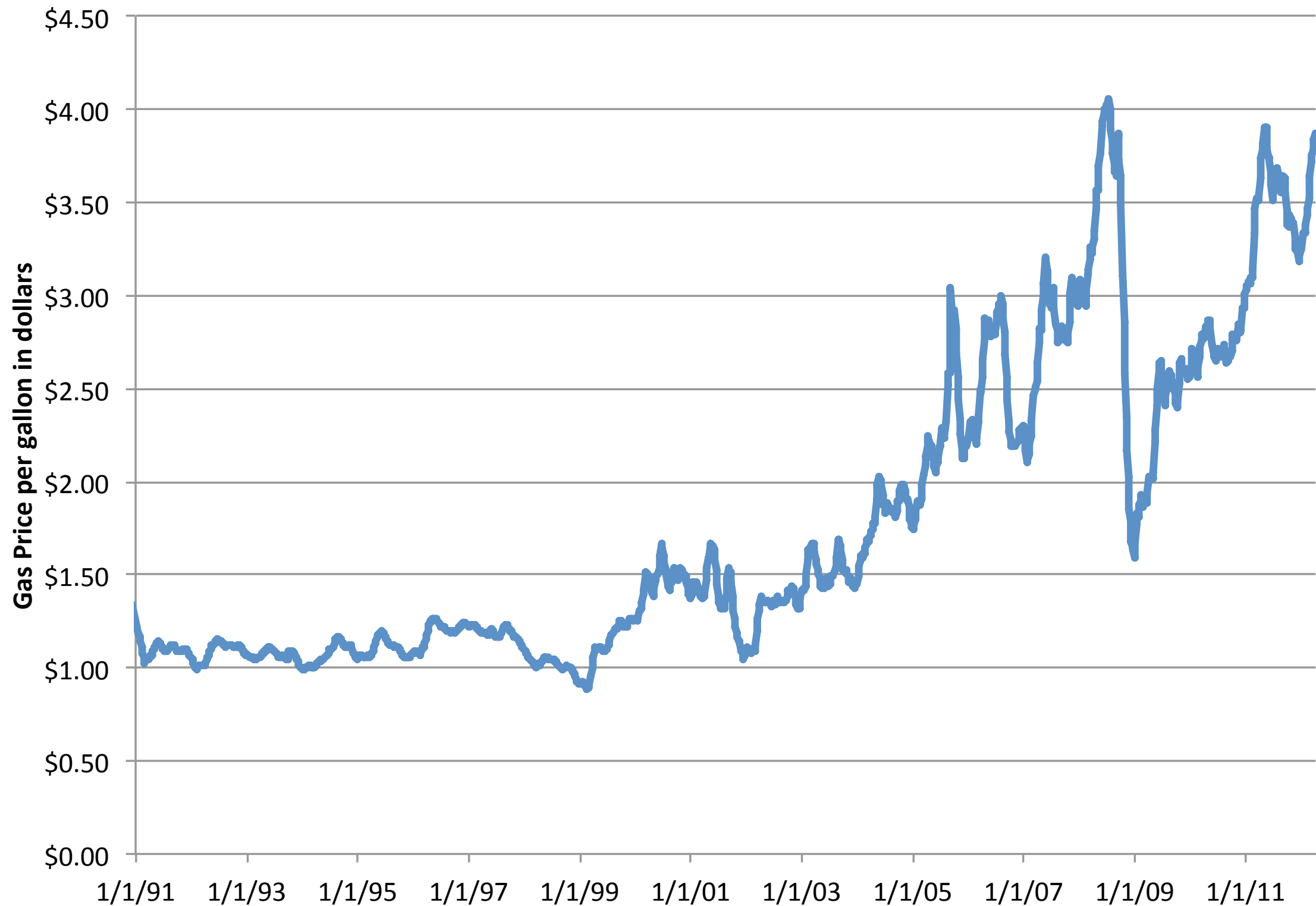
Constant dollars per barrel



Sources: Energy Information Administration (EIA) data; Bureau of Labor Statistics data.

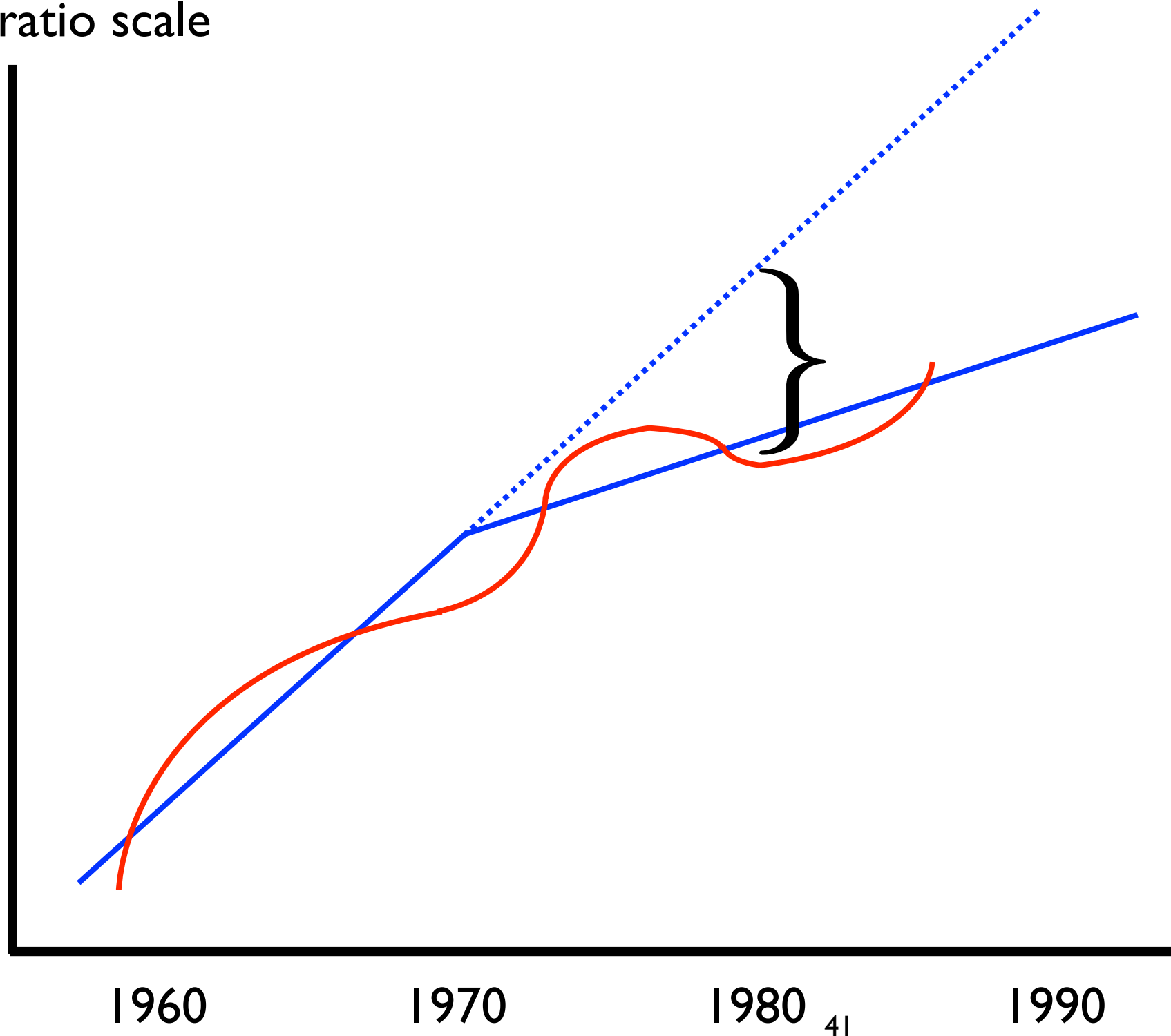
a. Monthly average price of West Texas Intermediate crude in dollars of November 2008.

Gas Prices: Do you remember 2008?



And something else happened

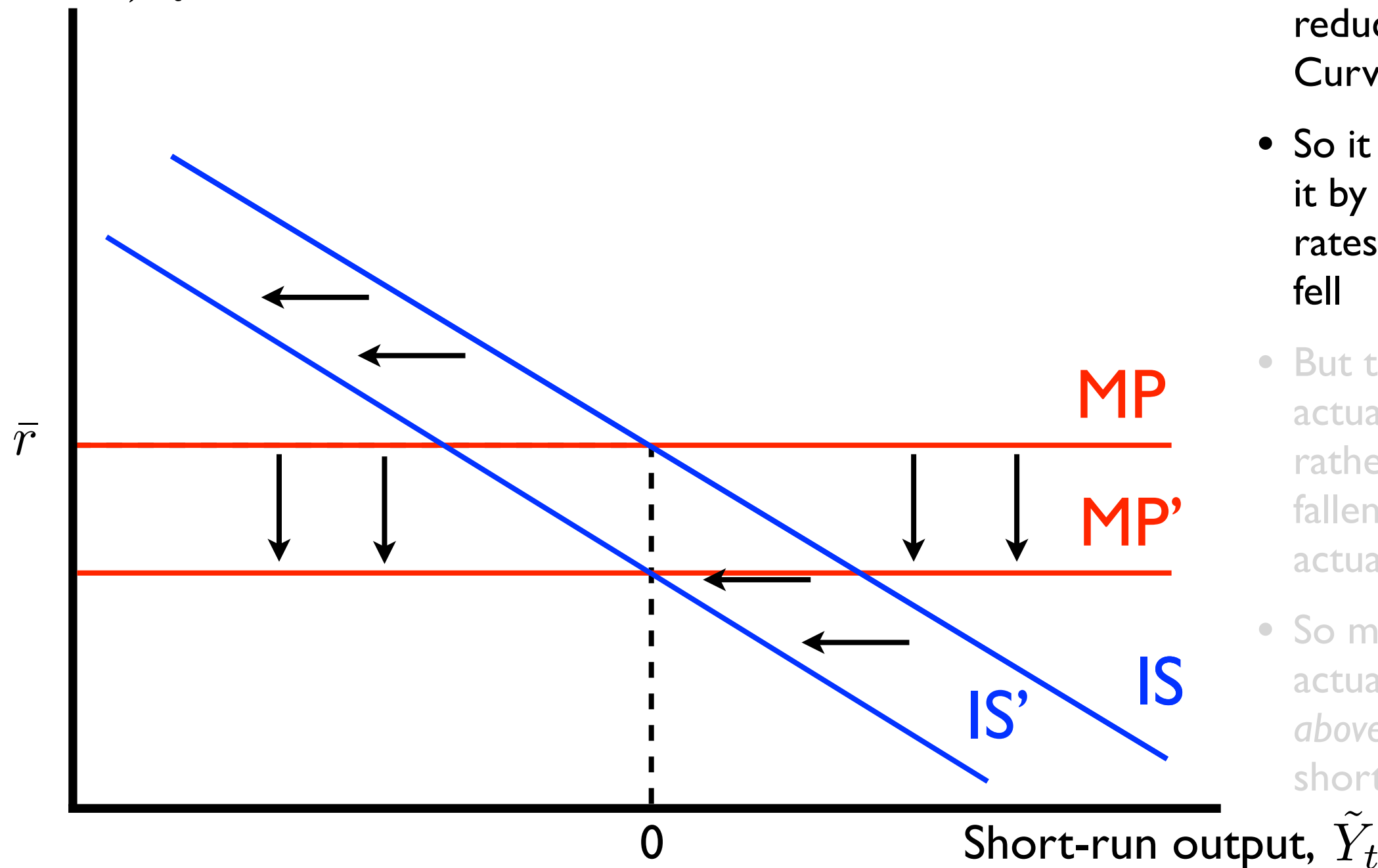
Real output,
ratio scale



- **Productivity growth slowed**, and so did growth in potential output!
- Visually, growth in potential was rapid from 1960–1970,
- And then a pretty big slowdown hit!
- We **know** this with 20/20 hindsight, but at the time, policymakers thought potential was continuing to grow
- Actual GDP was wiggling around,
- And relative to the old trend, this looked like a big recession!

Seeing what looked like a big recession, the Fed “loosened” monetary policy in response

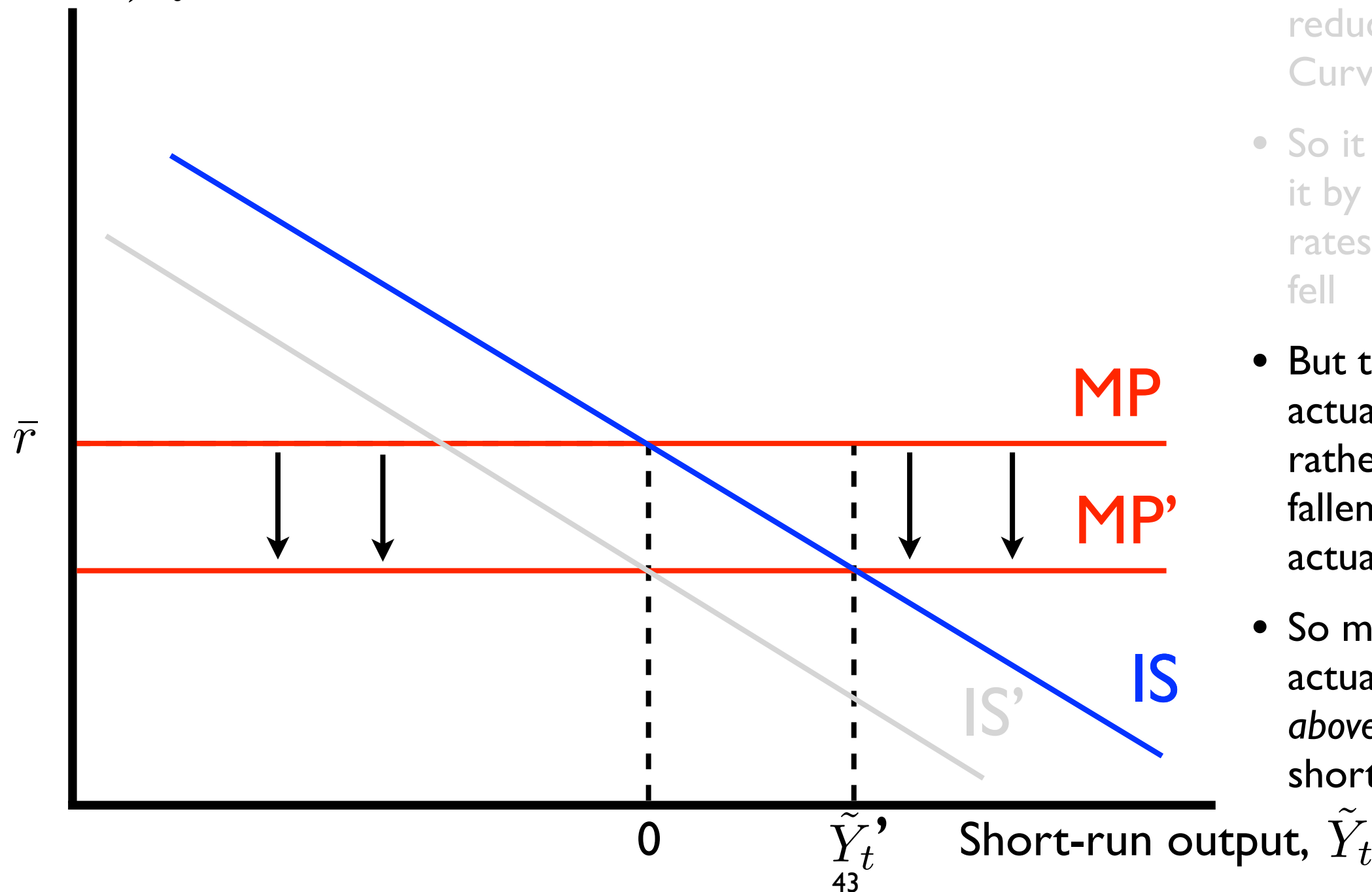
Real interest rate, R_t



- The Fed thought it saw a big recession, a reduction in the IS Curve
- So it tried to counteract it by lowering interest rates, and the MP curve fell
- But the economy wasn't actually in a recession; rather, potential has fallen. The IS Curve actually never moved
- So monetary policy is actually pushing output *above potential* in the short run!

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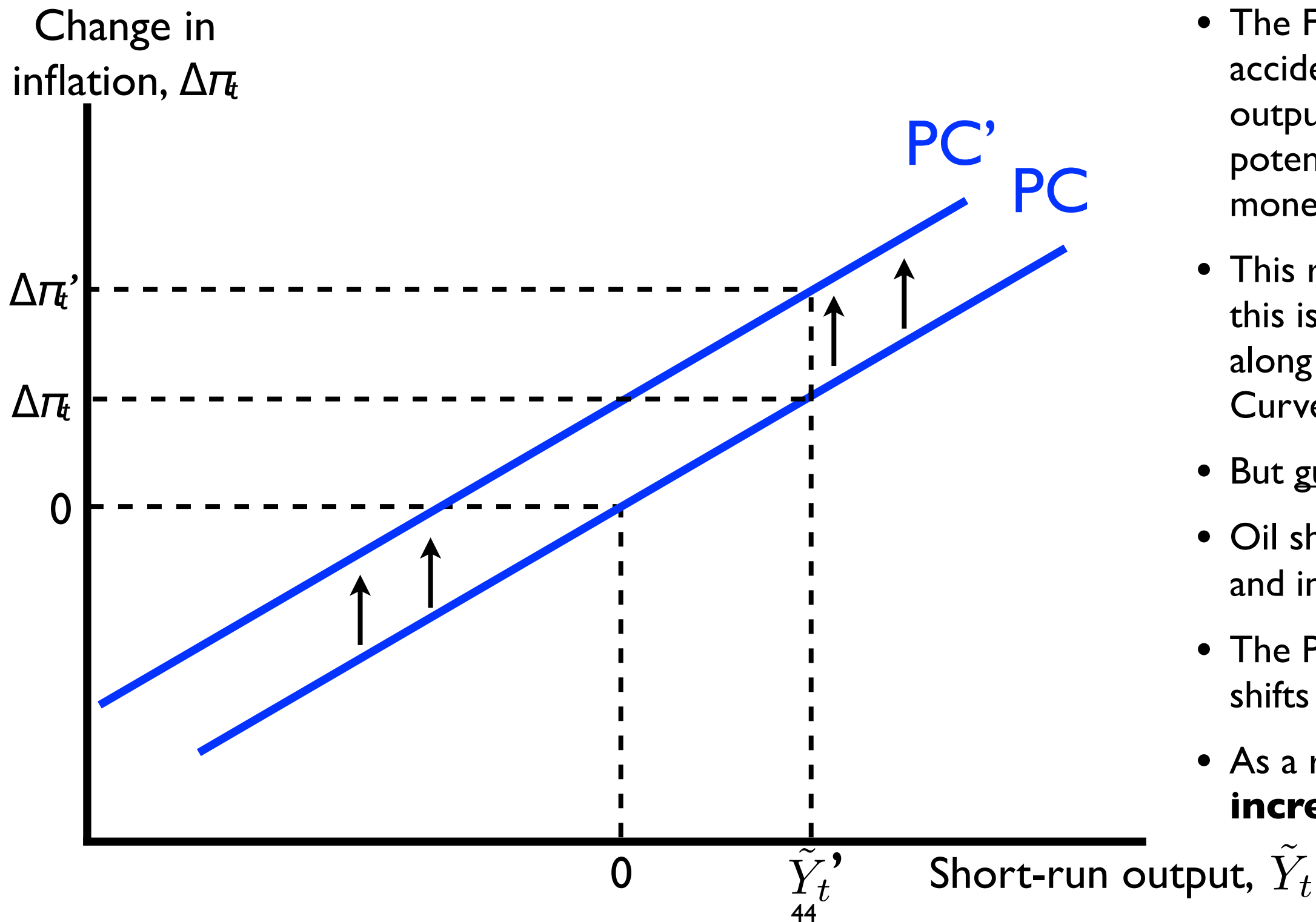
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This would be movement along the Phillips Curve

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$



- The Fed has accidentally stimulated output above potential with “easy” money
- This raises inflation; this is movement along the Phillips Curve
- But guess what else?
- Oil shocks also hit and increase $\bar{o}$
- The Phillips Curve shifts upward as well
- As a result, inflation **increases a lot**

What is the result of all this?

- This is typically called the period of “stagflation” — meaning **stagnant** growth in income per capita in addition to relatively rapid **inflation**
- Again, why did this happen? Several reasons:
 - There were oil shocks that raised prices of oil and other things
 - There was a reduction in growth of potential output, and the Fed mistook this for a recession, to be grown out of, and ran “easy” money policy with lower interest rates, stimulating the economy
 - Even if the Fed had perceived things correctly, they might not have believed they needed to trigger a *temporary* recession in order to reduce inflation from the oil shocks